

# Project Design Phase-II

## Solution Requirements (Functional & Non-Functional)

**Date:** 18 June 2026

**Team ID:** VIPC2-MERN-CHETHULA HARISH

**Project Name:** Book A Doctor – Online Doctor Appointment Booking System

---

## Functional Requirements

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)     |
|--------|-------------------------------|----------------------------------------|
| FR-1   | User Registration             | Registration through Registration Form |
|        |                               | User Profile Creation                  |
| FR-2   | User Authentication           | Login using Email and Password         |
|        |                               | Logout Functionality                   |
| FR-3   | Doctor Management             | Apply for Doctor Account               |
|        |                               | Admin Approval of Doctor Applications  |
| FR-4   | Doctor Search                 | View Approved Doctors                  |
|        |                               | Search Doctor Details                  |
| FR-5   | Appointment Management        | Book Appointment                       |
|        |                               | View Appointment Status                |
|        |                               | Cancel Appointment                     |
| FR-6   | Doctor Dashboard              | View Appointment Requests              |
|        |                               | Approve/Reject Appointments            |
| FR-7   | Notification Management       | Receive Appointment Notifications      |

|      |                  |                                       |
|------|------------------|---------------------------------------|
| FR-8 | Admin Management | Receive Doctor Approval Notifications |
|      |                  | Manage Users                          |
|      |                  | Manage Doctors                        |
|      |                  | Monitor System Activities             |

---

## Non-Functional Requirements

| NFR No. | Non-Functional Requirement | Description                                                                                                                   |
|---------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| NFR-1   | Usability                  | The system should provide a simple, user-friendly, and responsive interface for patients, doctors, and administrators.        |
| NFR-2   | Security                   | User passwords must be encrypted using bcryptjs, and secure authentication must be implemented using JWT.                     |
| NFR-3   | Reliability                | The application should consistently perform operations without data loss or system failures.                                  |
| NFR-4   | Performance                | System responses should be fast, and appointment operations should be processed efficiently.                                  |
| NFR-5   | Availability               | The application should be accessible online with minimal downtime.                                                            |
| NFR-6   | Scalability                | The system should support increasing numbers of users, doctors, and appointments without significant performance degradation. |

---

## Conclusion

The functional requirements define the core features required for the Book A Doctor platform, while the non-functional requirements ensure that the system remains secure, reliable, scalable, and user-friendly. Together, these requirements provide a strong foundation for successful system development and deployment.